AIN31306
Deep Learning

Predicting protein functions using deep learning

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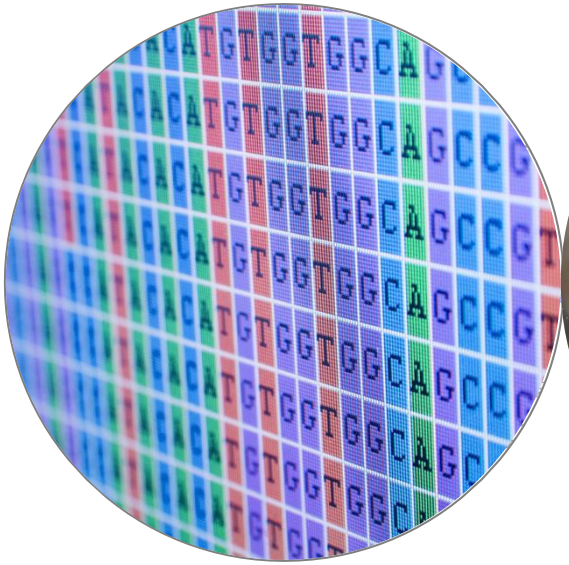
Originally by Dick de Ridder



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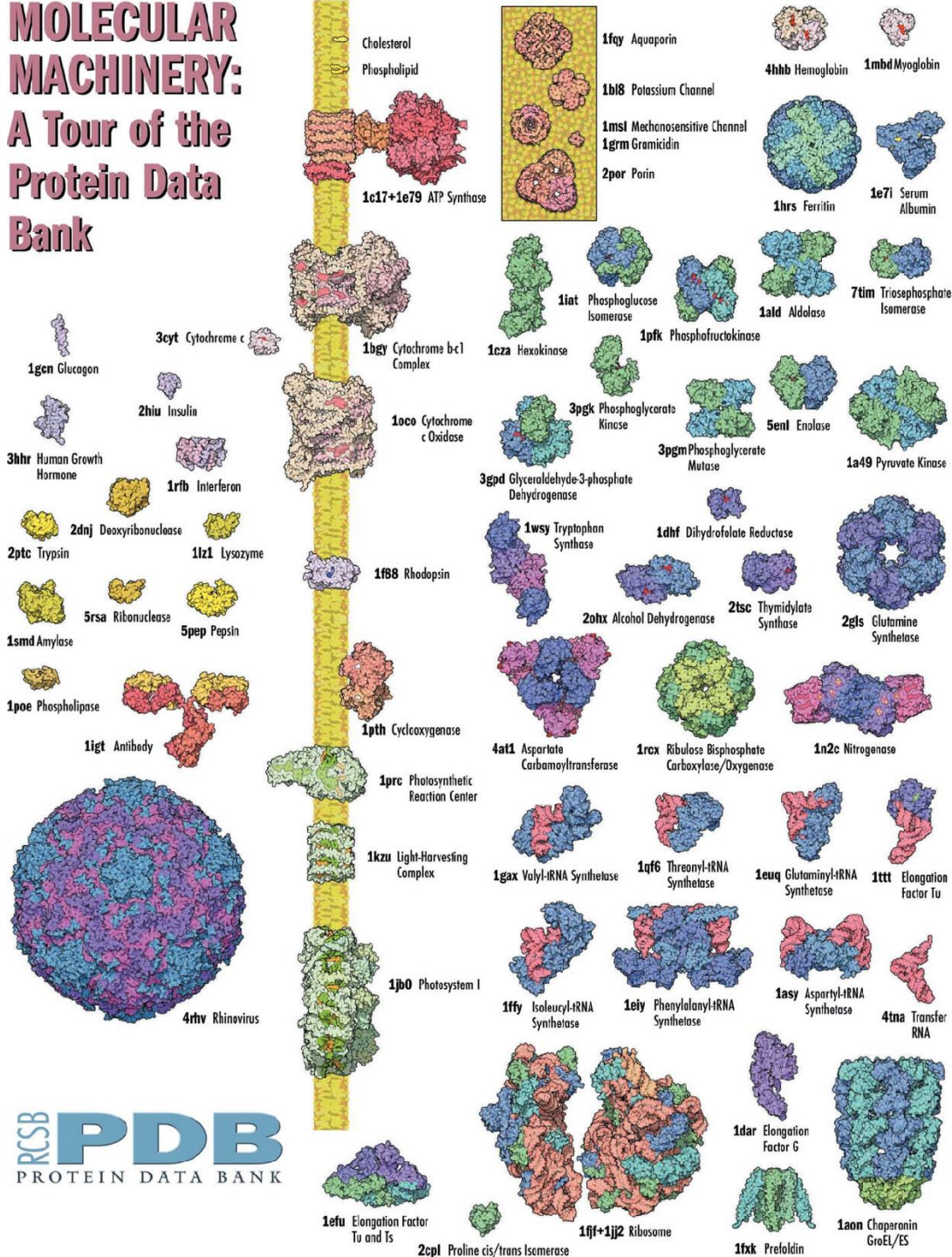
Proteins are the building blocks of life

- Sequence → Structure
- Structure → Function

- Measuring sequence is 'simple'

- Goal: predict function from sequence

- E.g.: AYG…VAGY…SK → Enzyme



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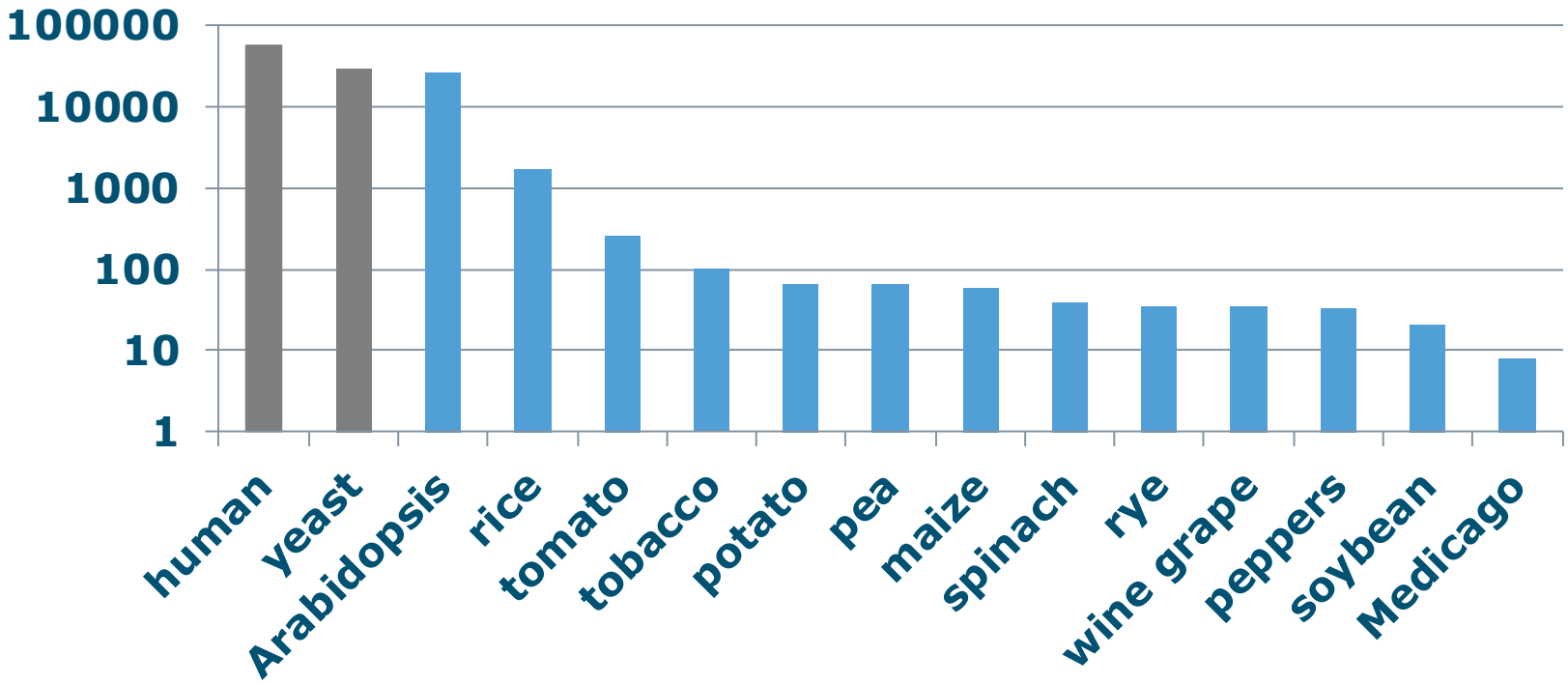
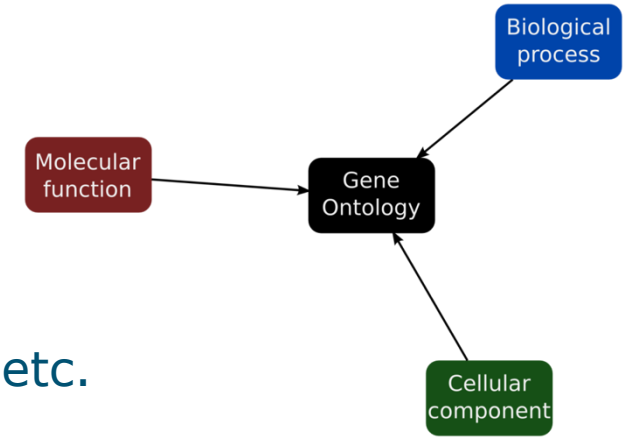
Experimental function annotation is scarce

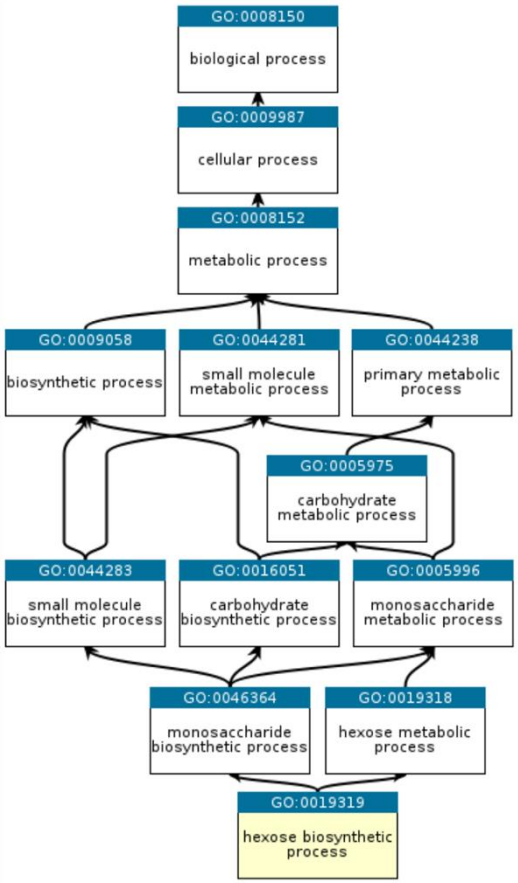


- Which proteins are involved in development, disease, etc.?
- How can we identify relevant genes to improve breeding?

The Gene Ontology



- Molecular function
  - E.g. catalytic activity, DNA binding, etc.
  - Directly related to sequence (e.g. domains)
- Biological process
  - Indirectly related to sequence
  - Dependent on biological context
- Cellular component
  - Where in the cell is the protein located?



Project data

- Simulated data
  - Six independent datasets (different level of difficulty)
  - Each dataset consists of sequences and corresponding annotation (which proteins perform “the function”)
- Human protein sequences (6,784)
  - Annotated for five GO terms of interest
- Final (unlabeled) test set of human proteins (14,765)

Project setup

- Basic structure to read and preprocess data is provided in handout - feel free to ignore, but at your own risk
- First develop model on simulated data, then apply without too many changes to human data
- Start with 1D CNN (we know this works well);
  if successful, explore fine-tuning a Transformer model
- **Checkpoint: if Thursday 16/4 at 9.00h not >75% accuracy on test set for at least one of the simulated datasets, e-mail rens.holmer@wur.nl and daniel.probst@wur.nl for discussion that day**